RIOT: An Operating System for the IoT

There are only a few operating systems that are suitable for Internet of Things (IoT) applications. RIOT OS, which is free and open source, is specially designed to meet the particular needs of the IoT, with features like a low memory footprint, high energy efficiency, real-time capabilities, a modular and configurable communication stack, and support for a wide range of low-power devices.

The Internet of Things (IoT) is used with heteregenous devices, which range from 8-bit to 32-bit microcontrollers from different manufacturers. Traditional operating systems or the conventional embedded OSs may not suit the requirements that these tiny devices demand (low power, a small size and low memory footprint). The RIOT operating system is a free and open source architecture that is friendly to IoT applications, and has proved to be a perfect OS for IoT systems.

Developed in 2008 as an OS for wireless sensor nodes, it was later fine tuned for IoT systems.

Features of RIOT

The OS is actively developed and maintained.

- There are no new programming environments. C or C++ can be used directly with existing tools like gcc, gdb, etc
- Less hardware dependent code
- Supports 8-,16- and 32-bit microcontroller platforms
- Energy efficieny is maintained
- Less interrupt latency, so real-time capability is ensured
- Multi-threading is enabled
- Supports the entire network stack of IoT (802.15.4 Zigbee, 6LoWPAN, ICMP6, Ipv6, RPL, CoAP, etc)
- Both static and dynamic memory allocation
- POSIX compliant (partial)
- All output can be seen in the terminal if hardware is not available; however, there is a visualisation tool called RIOT-TV that is provided

Architecture, board and driver support

It supports various architectures like

- MSP430
- ARM7
- Cortex-M0, M3 and M4
- x86, etc

It also supports the native port, where one can simulate the output within the OS it is running. So RIOT is supported in Linux as well as OS X.

There are in-built drivers for the following sensors (without the need for hardware, these sensors can be modelled in native mode also):

- Radio receivers
- Environmental sensors for humidity, temperature, pressure, alcohol, gas, etc
- Accelerometers
- Gyroscopes
- Ultrasonic sensors, light and servo motors

Most of the sensor boards like TelosB, ST, Zolertia, MSP 430, Arduino and Atmel have support from this OS (and the list is quite long). RIOT also supports virtualisation, where the code and application can run as a simple UNIX process. It also uses Wireshark for packet sniffing.